

DATA & BUSINESS INTELLIGENCE

A portfolio of five analytical projects

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Four operational projects use synthetic data. The Tableau dashboard uses a historical public dataset. Findings are demonstrations, not claimed employer savings.

01 / MySQL / Python / Excel

Compliance Reporting & Data Quality

Which cases need attention, and is the latest report reliable?

266

open cases

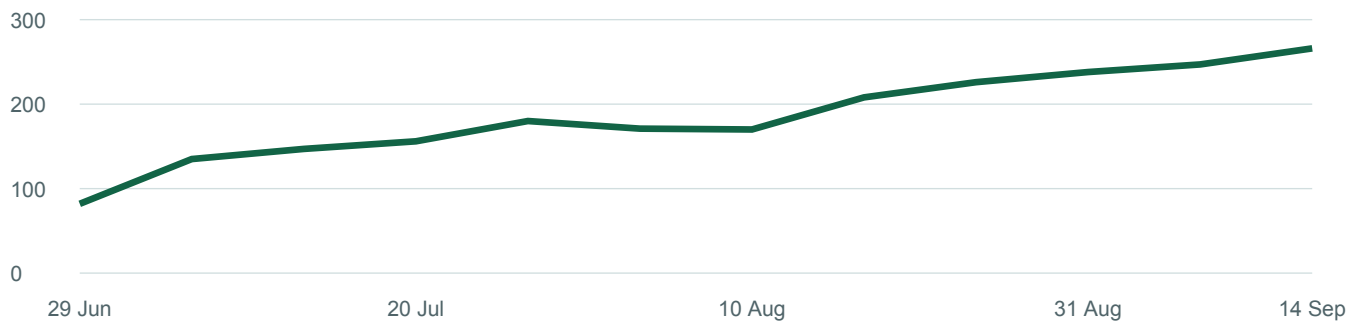
157

overdue open

33.9%

on-time due cohort

Weekly closing backlog (cases; week starting)



Built: MySQL department joins and daily/weekly/monthly summaries; an Excel report with a native PivotTable and case register; Python validation and publication controls.

Finding: 59.0% of the open backlog is overdue. Warehouse has 99 open cases, including 56 overdue. Review ageing, case mix and ownership before drawing conclusions about performance.

Evidence: All headline, department and 100 period-row outputs reconcile to independent calculations. Nineteen tests pass. Invalid and stale inputs preserve the last successful report; late valid runs are flagged.

Scope: 29 June-20 September 2026. Partial June/September months. The generator deliberately slows some Warehouse cases. No production scheduler or email service; desktop Excel refresh remains unverified.

[Open case study, workbook and source code](#)

DELIVERY & INVENTORY

Turning operational records into reviewable decisions

02 / Power BI / DAX / Relational modelling

Supplier Delivery Performance

Which delivery commitments are being missed?

38.6%

on time & in full

18

overdue open lines

18.0 days

mean received lead time

Built: A two-page Power BI report with four related tables, eight DAX measures and supplier/category/month filters. OTIF includes overdue unreceived lines in the 1,488-line due population.

Finding and action: 575 due lines arrived on time and in full. Review overdue and short-received lines with suppliers to establish causes and recovery dates.

Evidence and scope: Supplied Windows validation records a successful refresh and filter test. Synthetic snapshot, 21 September 2026. One receipt per line; no live ERP or scheduled cloud refresh. The case page links to the actual PBIX and model source.

03 / Excel / VLOOKUP / SUMIFS / COUNTIFS

Inventory Health & Replenishment

What needs replenishing, and what would it cost?

27 / 80

items flagged

£766,792

proposed purchase cost

£2.18m

on-hand stock value

Built: Exact-match VLOOKUPs and SUMIFS/COUNTIFS connect stock, demand and lead-time assumptions. The workbook calculates stock position, reorder points, proposed quantities and category purchasing cost.

Finding and action: 27 items require review, with sinks accounting for about £290,239 of the proposed spend. Confirm allocations, receipt dates and purchasing constraints before ordering.

Evidence and scope: 80-item calculation model with Tables, validation, conditional formatting and a native chart. No pack sizes, minimum order quantities or seasonality. £766,792 is a scenario estimate, not an approved order or saving. Desktop Excel refresh remains unverified.

DATA QUALITY & EXPLORATION

Making assumptions and failure behaviour visible

04 / Python / Validation / Unit tests / BOM checks

Supply-Chain Reporting Controls

Can bad data be stopped before a report is replaced?

12

tests passed

9

issues detected

5

dirty rows rejected

Built: A Python pipeline validating purchase-order keys, dates, quantities, item/supplier references and BOM cycles. CSV diagnostics, JSON summaries and HTML reports retain an audit trail.

Finding and action: The 1,501-row dirty fixture produces nine issues and five rejected rows. Failed validation leaves the last successful report pointer unchanged. Correct the source and rerun before publication.

Evidence and scope: Twelve tests pass. Clean and blocked HTML reports are available from the case page. Local implementation only; Power Automate delivery is a proposed workflow, with no deployed scheduler or email integration.

05 / Tableau / Calculated fields / Interactive filters

Netflix Content Portfolio

How does the catalogue differ by genre, rating and geography?

8,780

source titles

4

analytical questions

Tableau Public

interactive dashboard

Built: An interactive Tableau Public dashboard comparing country fields, genre composition, audience ratings and movie-only runtime, with release-year and director filters.

Interpretation: Explore differences in catalogue composition; do not equate title counts with demand, engagement or profitability. Normalising multi-country and multi-genre fields would improve more detailed geographical analysis.

Evidence and scope: 8,780 titles in the connected worksheet. Movie runtime excludes TV-season values; missing countries/directors remain visible. A historical public dataset, not the current Netflix catalogue.

[Open the interactive Tableau dashboard](#)